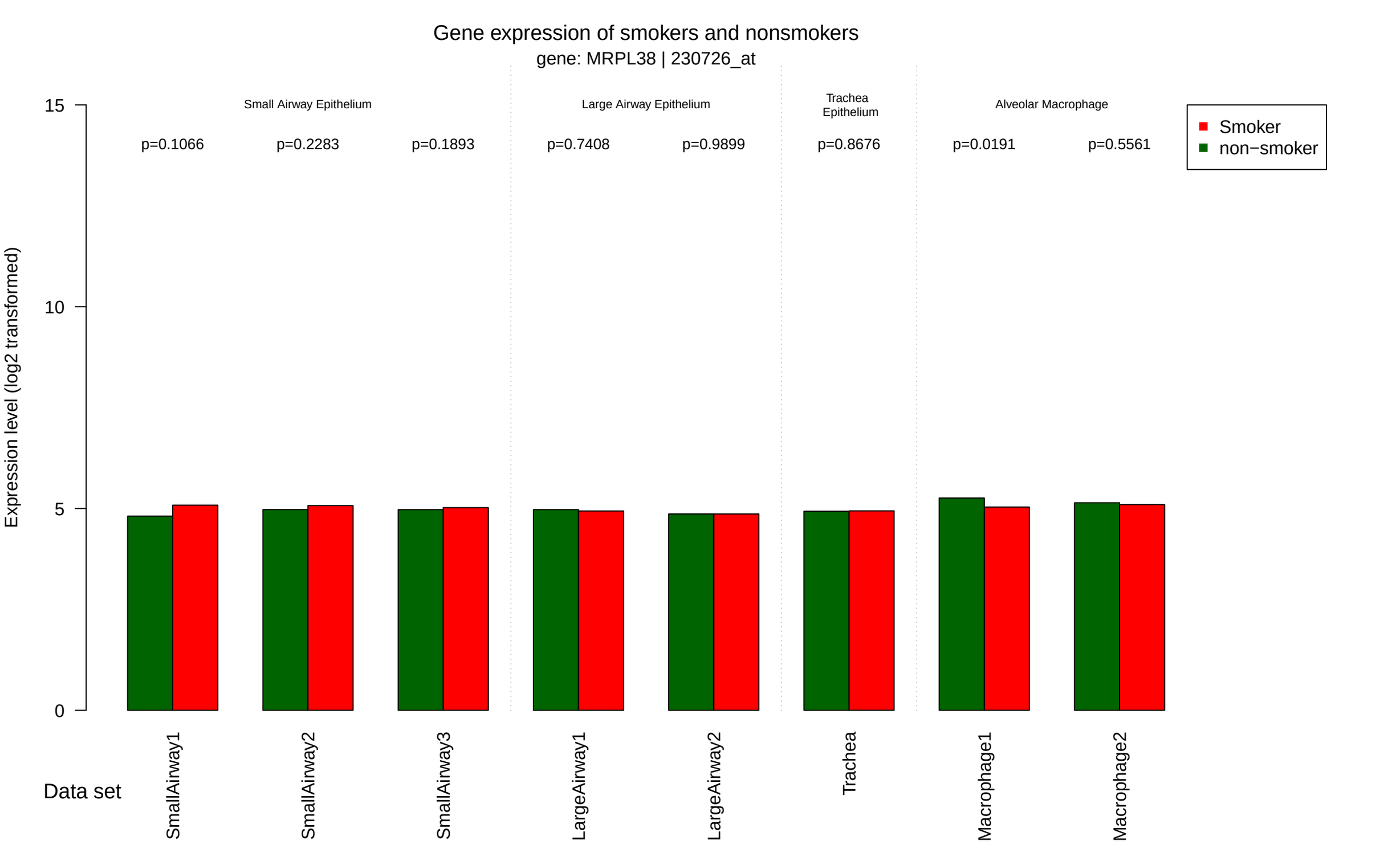
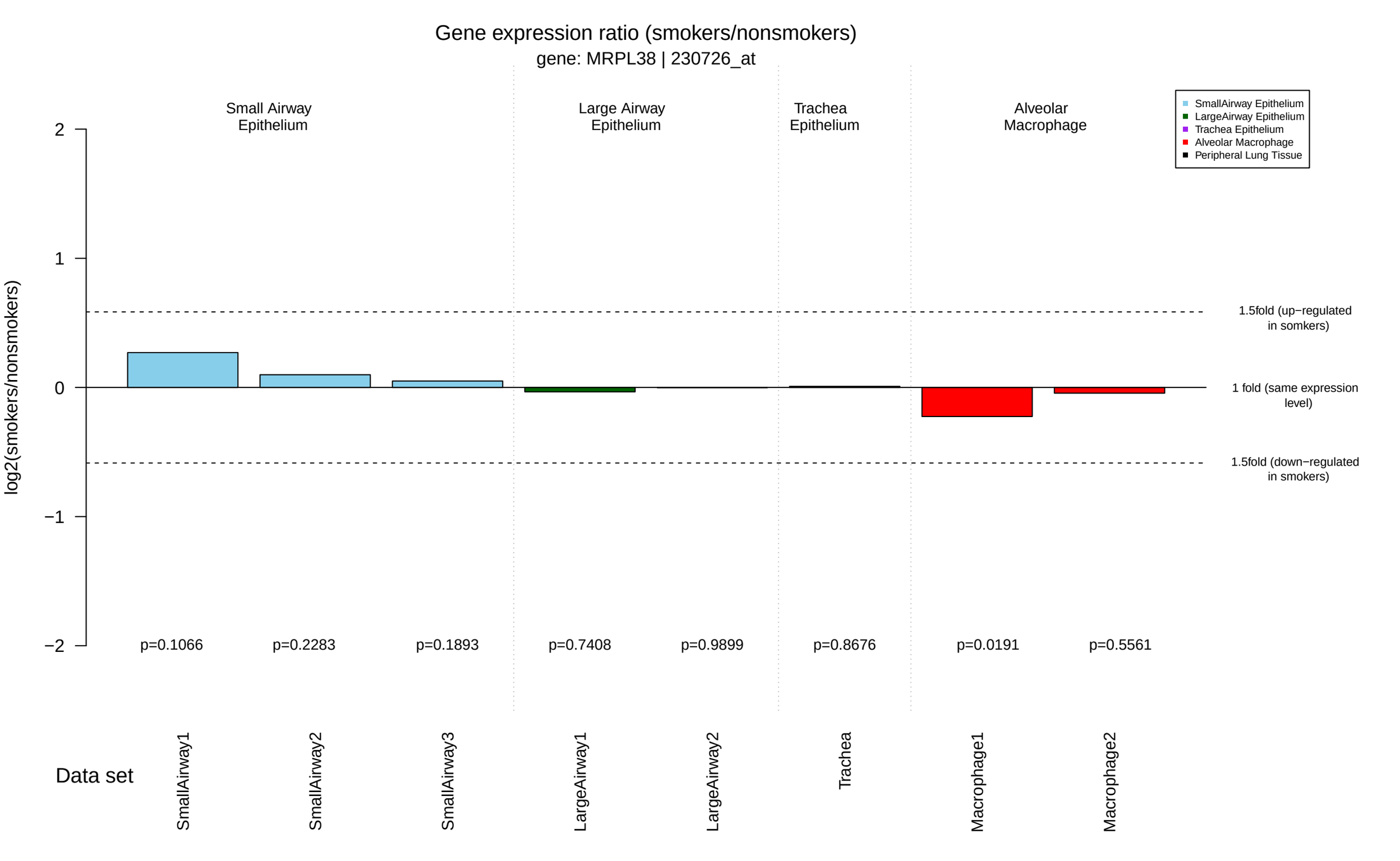
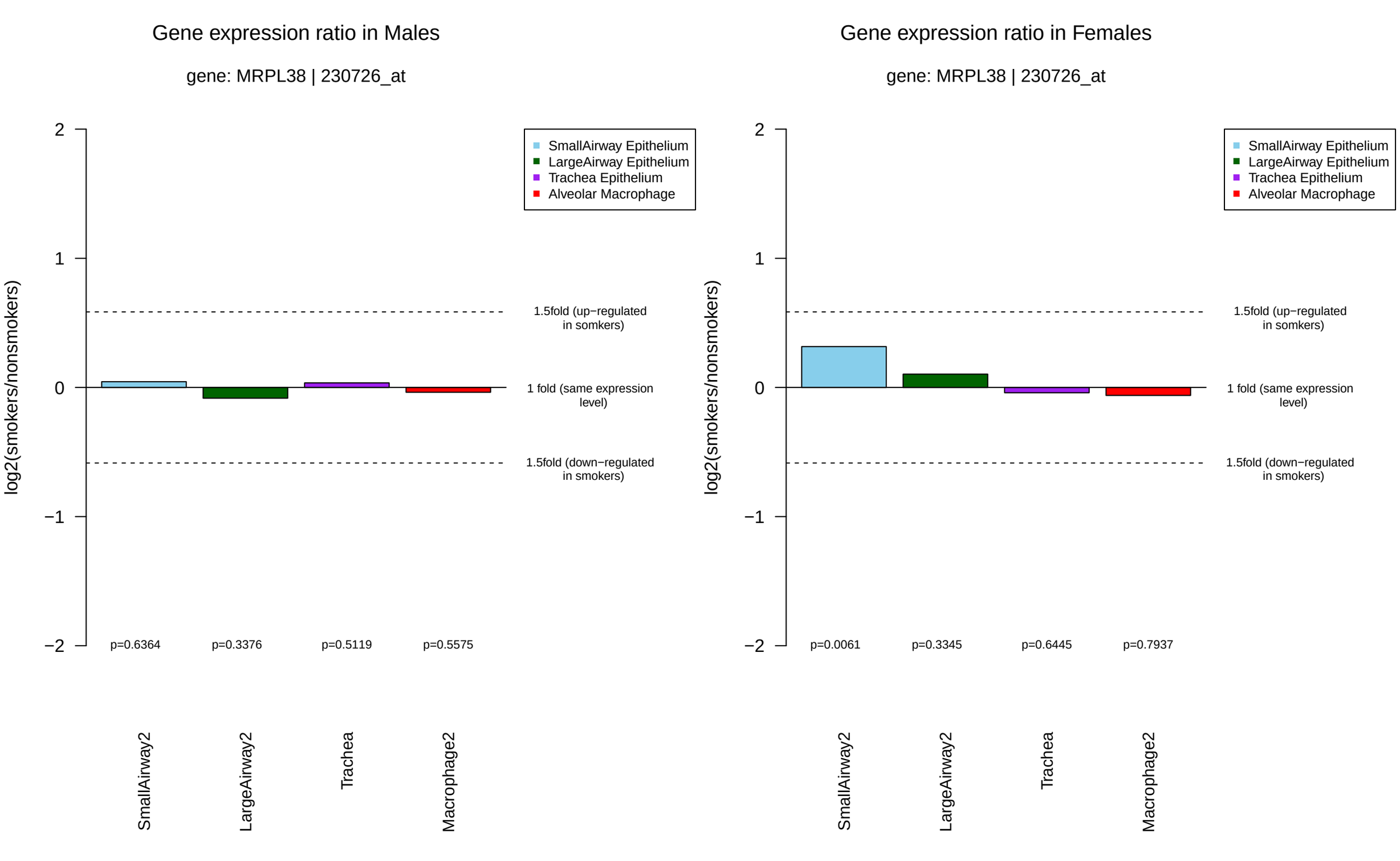
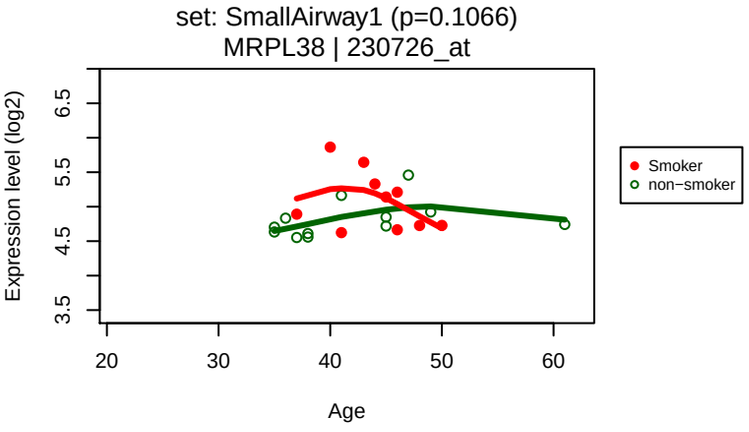
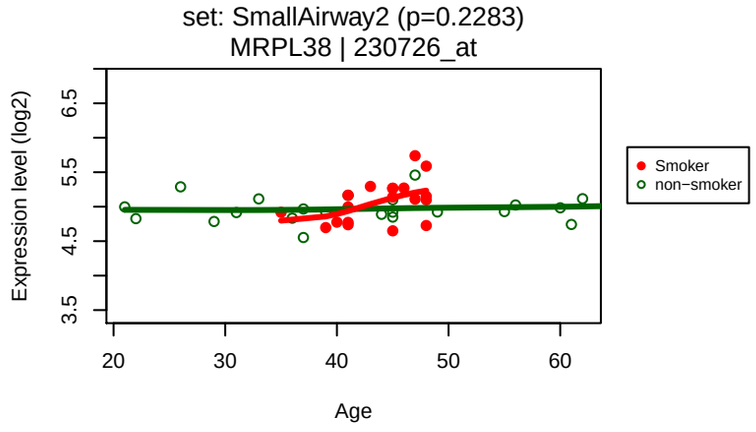
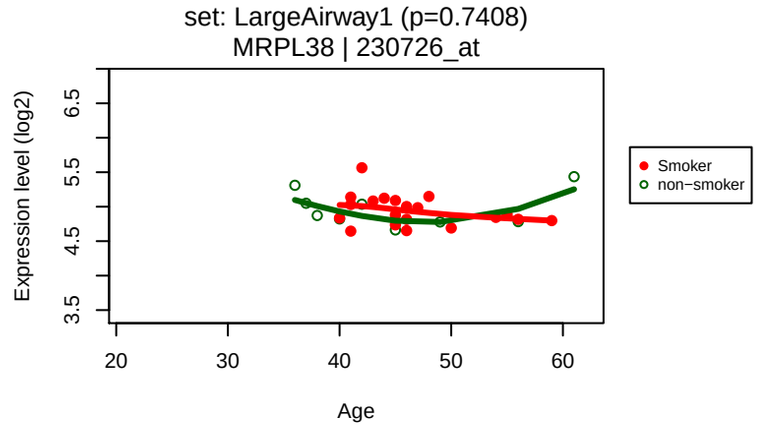

Gene expression of smokers and nonsmokers
gene: MRPL38 | 230726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1066
p=0.2283
p=0.1893
p=0.7408
p=0.9899
p=0.8676
p=0.0191
p=0.5561
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

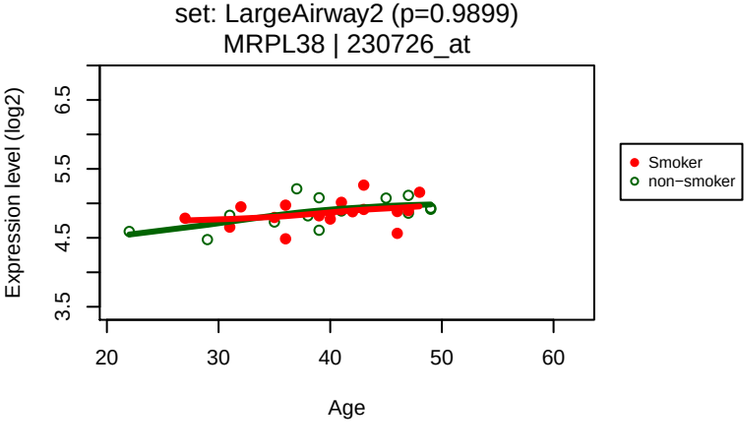

Gene expression ratio (smokers/nonsmokers)
gene: MRPL38 | 230726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1066
p=0.2283
p=0.1893
p=0.7408
p=0.9899
p=0.8676
p=0.0191
p=0.5561
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

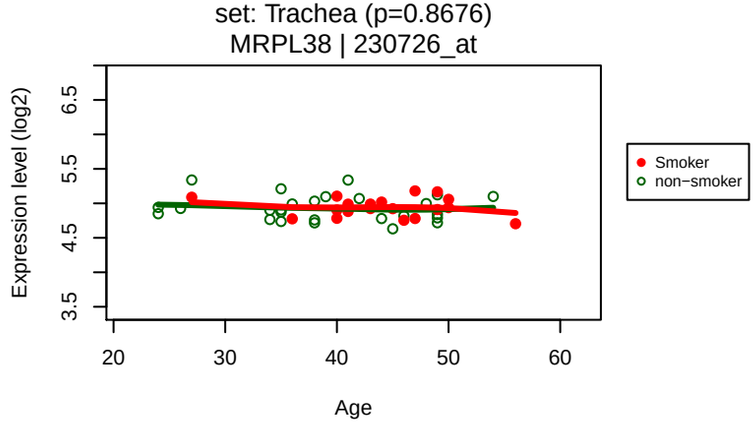

Gene expression ratio in Males
gene: MRPL38 | 230726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6364
p=0.3376
p=0.5119
p=0.5575
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL38 | 230726_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0061
p=0.3345
p=0.6445
p=0.7937
SmallAirway2
LargeAirway2
Trachea
Macrophage2

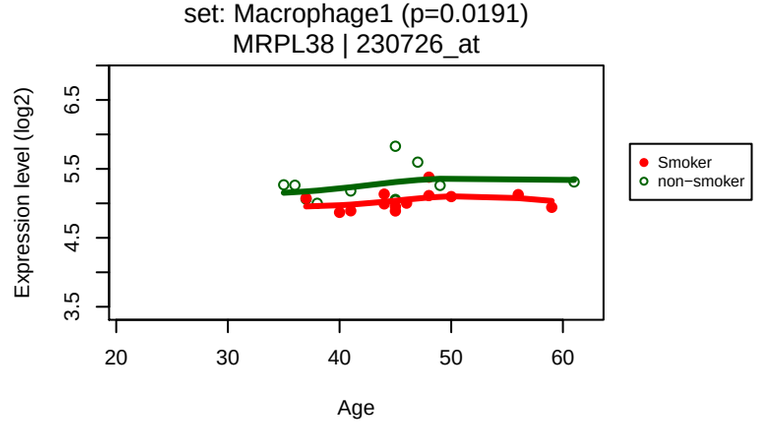

set: SmallAirway1 (p=0.1066)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

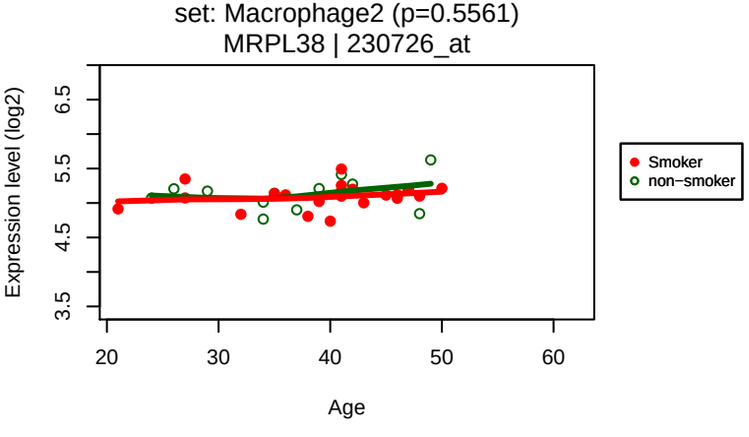

set: SmallAirway2 (p=0.2283)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker

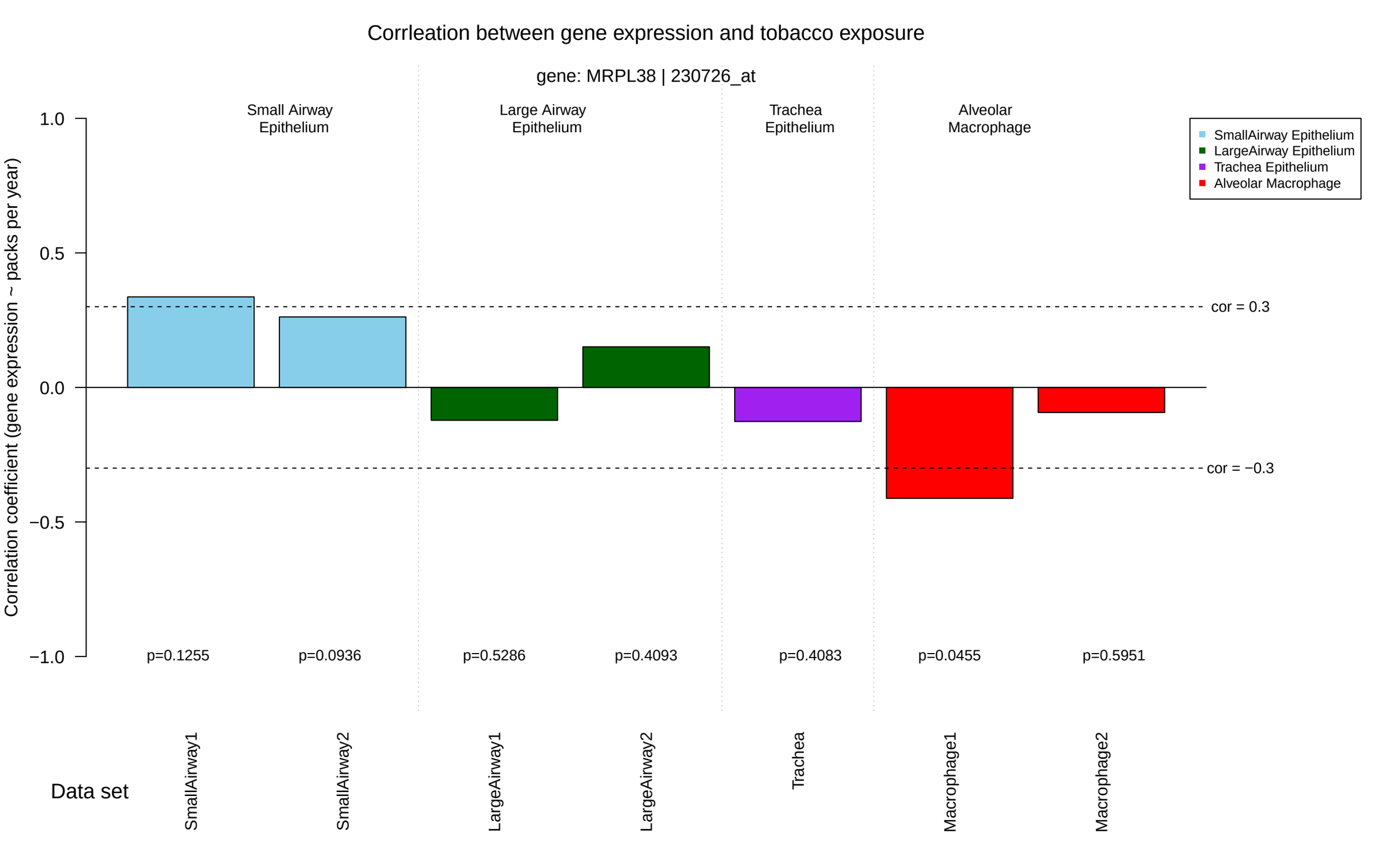

set: LargeAirway1 (p=0.7408)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9899)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.8676)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0191)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5561)
MRPL38 | 230726_at
Expression level (log2)
3.5
4.5
5.5
6.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL38 | 230726_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1255
p=0.0936
p=0.5286
p=0.4093
p=0.4083
p=0.0455
p=0.5951
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2